

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our recent predictive analysis has established a highly reliable model, capable of accurately forecasting outcomes with over 83% precision. This strong performance provides a robust foundation for strategic planning and confidently supports our ability to identify key influencing factors. The consistent accuracy ensures that the insights derived are trustworthy and actionable for business decision-making. The analysis clearly indicates that demographic characteristics and financial investment are the dominant forces shaping the predicted outcome. Specifically, gender emerges as the most influential factor, closely followed by an individual's age and the fare paid. This highlights that an individual's profile, particularly their gender, alongside their financial commitment, plays a critical role in determining the likelihood of the event. Given the reliability of these insights, we recommend a focused strategy of monitoring these key demographic and economic drivers. Proactively tracking changes in gender distribution, age profiles, and fare structures will be essential for anticipating shifts in outcomes and enabling more targeted, effective business interventions. These insights offer a strong basis for optimizing strategies and resource allocation.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

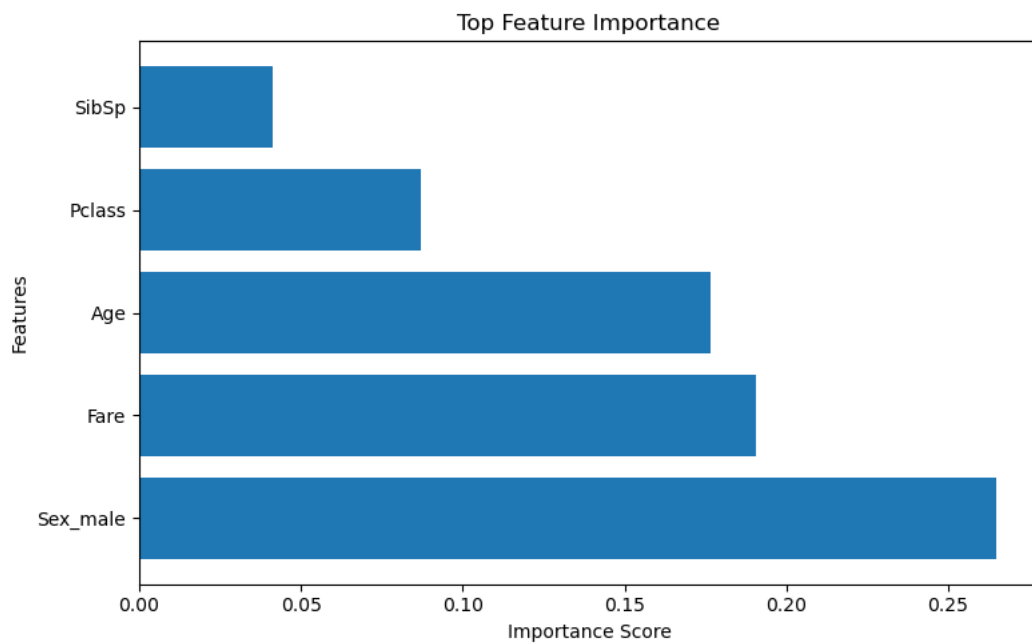
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

